IPv6
ARP,RARP, BOOTP,DHCP

Dr. Gunjan Chugh
Assistant Professor
DTU

IPv6 Addressing

Introduction to IPv6

- IPv6 stands for Internet Protocol Version 6
- Designed to replace IPv4
- Developed to solve IPv4 address exhaustion
- Provides larger address space and improved efficiency

IPv6 Address Size

- IPv4 = 32 bits
- IPv6 = **128 bits**

Total possible addresses ≈ 340 undecillion

Why IPv6 is Needed

- IPv4 uses 32-bit addressing (~4.3 billion addresses)
- Rapid growth of internet devices (IoT, smartphones, cloud systems)
- IPv6 uses 128-bit addressing
- Provides almost unlimited address space

Types of IPv6 Addresses

- Unicast
  - One sender to one receiver
- Multicast
  - One sender to multiple receivers
- Anycast
  - One sender to nearest receiver in a group
- Note: IPv6 does NOT support broadcast.

IPv6 Address Format

- IPv6 address characteristics:
- Length: **128 bits**
- Written in **hexadecimal (base 16)**
- Divided into **8 groups**
- Each group contains **4 hexadecimal digits**
- Groups separated by colons (:)

Example:

- 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Structure of IPv6 Address

- An IPv6 address is divided into two main parts:
- Network Prefix
  - Identifies the network
  - Similar to network ID in IPv4
- Interface Identifier
  - Identifies the device (host)
  - Similar to host ID

Typical format:

- Network Prefix (first 64 bits)
  Interface ID (last 64 bits)

IPv6 Address Compression Rules

- IPv6 allows shortening of addresses using two rules:
- Rule 1: Remove Leading Zeros
  Example:
  0db8 → db8
  0001 → 1
- Rule 2: Replace Consecutive Zeros with Double Colon (::)
  Example:
  2001:0db8:0000:0000:0000:0000:1428:57ab
  Can be written as:
  2001:db8::1428:57ab
- Important Rule:
  Double colon (::) can be used only once in an address.

Key Features of IPv6

- 128-bit Address Space
  - Extremely large number of addresses
- Simplified Header
  - Fixed size (40 bytes)
- No NAT Required
  - Each device can have unique public IP
- Built-in Security
  - IPsec support

Key Features of IPv6

- No Broadcast
  - Uses multicast instead
- Better Routing Efficiency
  - Hierarchical addressing
- Improved QoS Support
  - Traffic Class and Flow Label fields

PACKET FORMAT

The IPv6 packet is shown in Figure. Each packet is composed of a mandatory base header followed by the payload. The payload consists of two parts: optional extension headers and data from an upper layer. The base header occupies 40 bytes, whereas the extension headers and data from the upper layer contain up to 65,535 bytes of information.

Figure *IPv6 datagram*

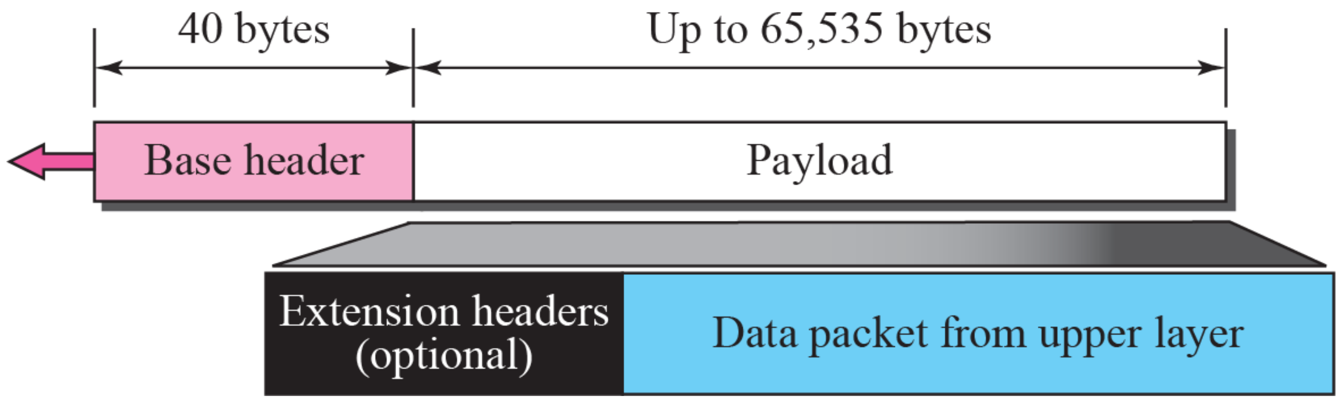


Figure *Format of the base header*

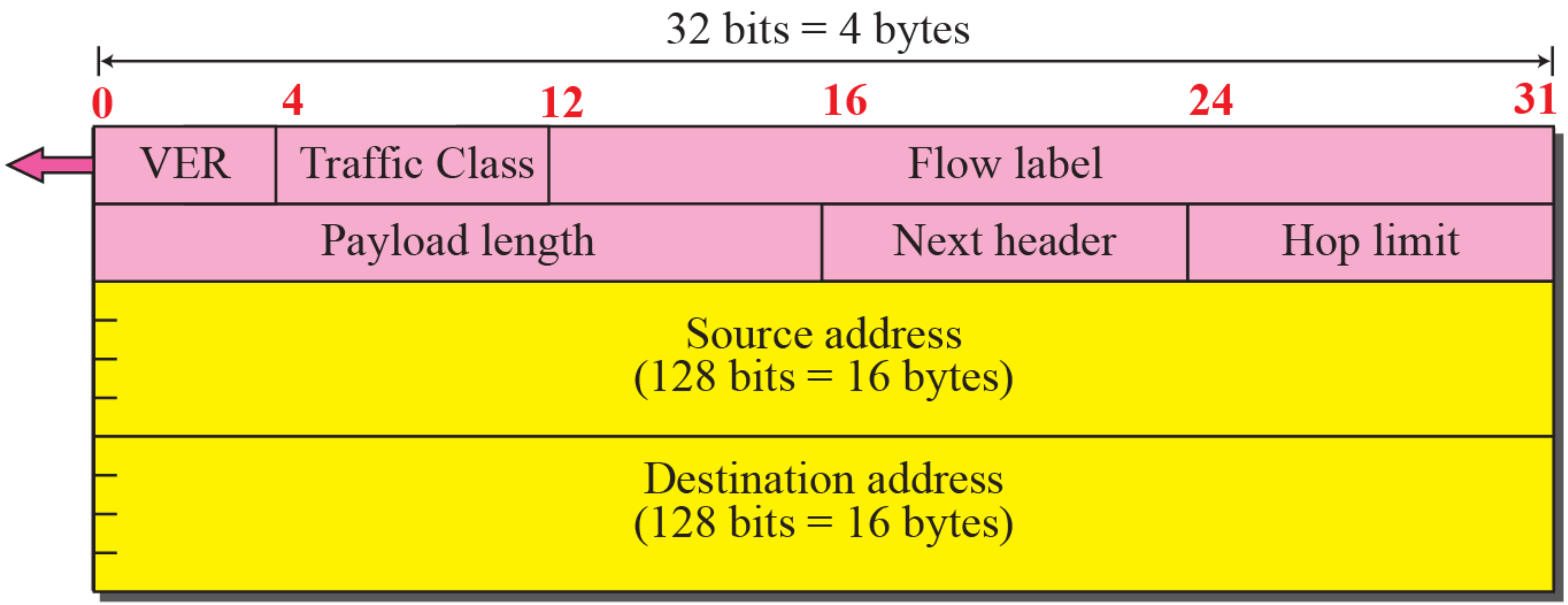
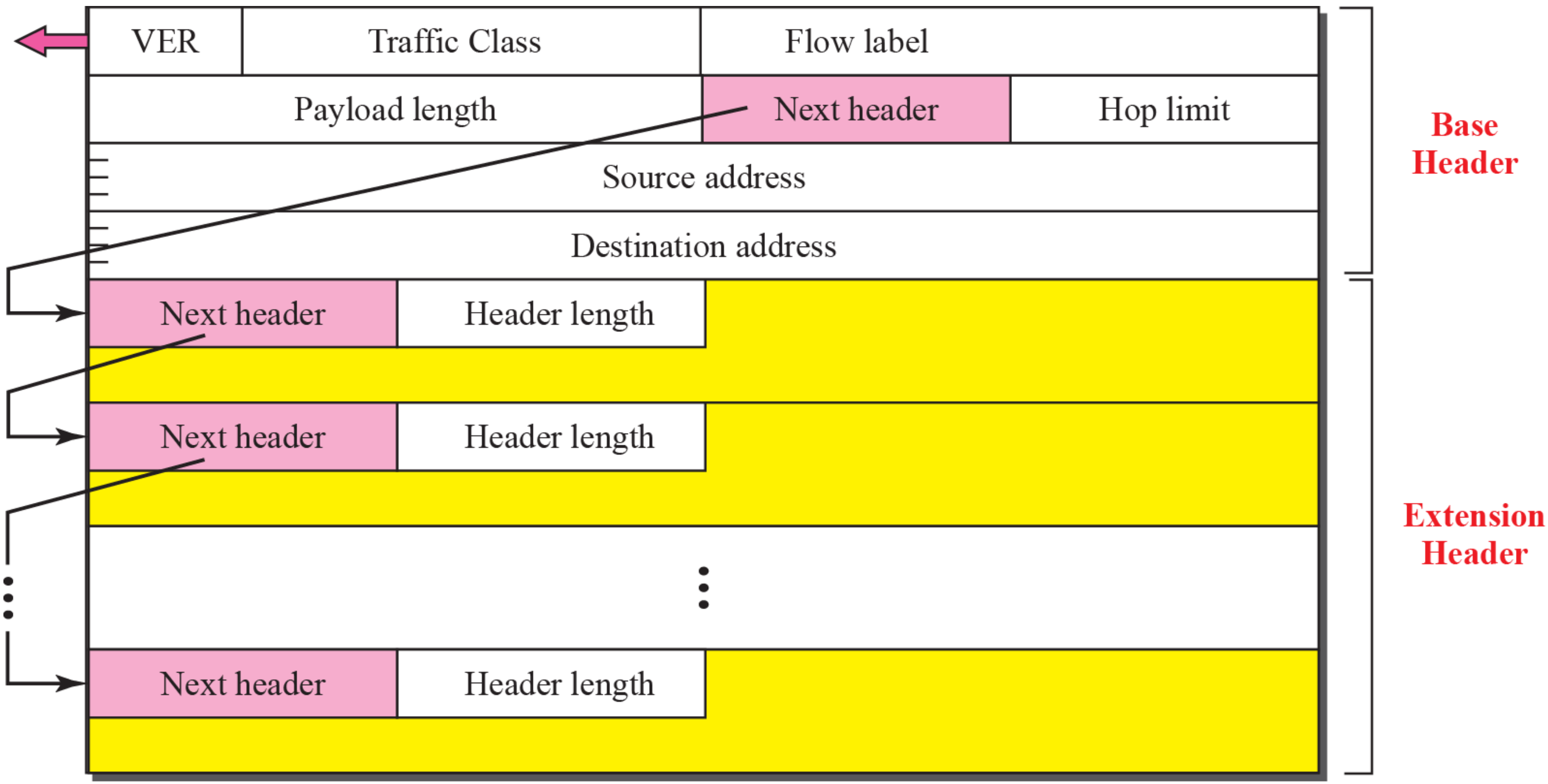


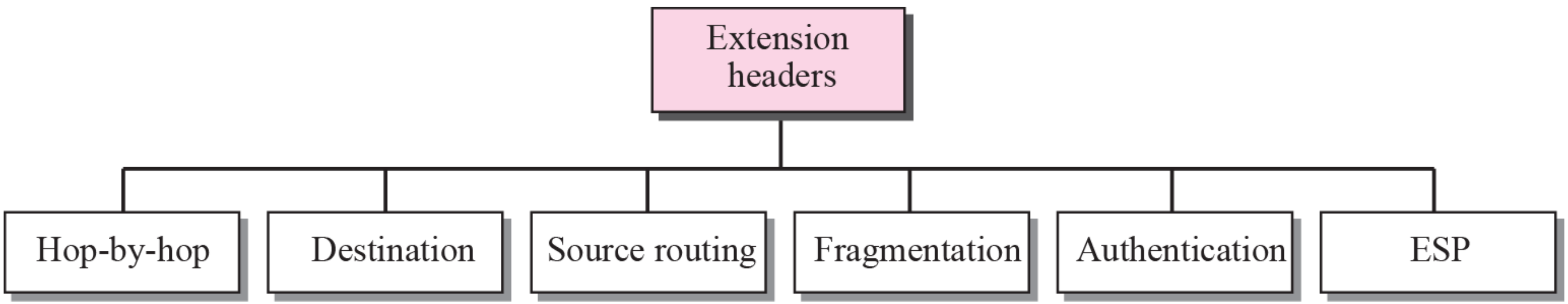
Figure *Extension header format*



IPv6 Header Fields

Field	Size	Explanation
Version	4 bits	Identifies the IP version. Always set to 6 for IPv6 packets.
Traffic Class	8 bits	Indicates packet priority and QoS handling (similar to ToS in IPv4).
Flow Label	20 bits	Identifies packets belonging to the same flow (used for real-time traffic).
Payload Length	16 bits	Specifies the size of data after the IPv6 header (in bytes).
Next Header	8 bits	Indicates the type of header following IPv6 (TCP, UDP, ICMPv6, or extension header).
Hop Limit	8 bits	Limits packet lifetime; decremented at each router (like TTL in IPv4).
Source Address	128 bits	IPv6 address of the sender.
Destination Address	128 bits	IPv6 address of the receiver.

Figure *Extension header types*



Extension Header

Extension Header	**Purpose**
Hop-by-Hop Options Header	Processed by every router along the path; used for special options like jumbo payload.
Routing Header	Specifies a list of intermediate nodes the packet must pass through.
Fragment Header	Used when sender fragments the packet; routers do not fragment in IPv6.
Destination Options Header	Contains options processed only by the destination node.
Authentication Header (AH)	Provides integrity and authentication (used in IPsec).
Encapsulating Security Payload (ESP)	Provides encryption and security for packet data (IPsec).

Advantages of IPv6

- Solves IPv4 address exhaustion
- Supports IoT and future technologies
- More efficient packet processing
- Better support for real-time applications
- Enhanced scalability

Special IPv6 Addresses

Address Type	Example	Meaning	Where Used	Routable on Internet?	Memory Trick
Unspecified	::	No address assigned	During startup / requesting IP	No	"I don't know my address yet"
Loopback	::1	Self-communication	Testing local machine	No	"Talking to myself"
Link-Local	FE80::	Local network communication	Same LAN only	No	"Inside my classroom only"
Global Unicast	2000::	Public IPv6 address	Internet communication	Yes	"My official home address"

Comparison: IPv4 vs IPv6

Parameter	IPv4	IPv6
Address Length	32 bits	128 bits
Address Format	Decimal (e.g., 192.168.1.1)	Hexadecimal (e.g., 2001:db8::1)
Address Space	~4.3 billion	Extremely large (almost unlimited)
Header Size	Variable (20–60 bytes)	Fixed (40 bytes)
Checksum Field	Present	Removed
Fragmentation	Done by sender and routers	Done only by sender
Broadcast	Supported	Not supported
Multicast	Supported	Supported (replaces broadcast)
Anycast	Not supported	Supported
Configuration	Manual / DHCP	SLAAC / DHCPv6
Security	Optional (IPsec)	Built-in IPsec support
NAT Requirement	Required due to limited IPs	Not required
QoS Support	Limited	Improved (Traffic Class + Flow Label)

Address Resolution Protocol (ARP)

Address Mapping

- The delivery of a packet to a host or a router requires two levels of addressing: *logical* and *physical*.
- We need to be able to map a logical address to its corresponding physical address and vice versa.
- These can be done using either *static* or *dynamic* mapping.

Address Mapping

- Anytime a host or a router has an IP datagram to send to another host or router, it has the **logical (IP) address of the receiver.**

- But the IP datagram must be encapsulated in a frame to be able to pass through the physical network.

- **The sender needs the physical address of the receiver.**

- A mapping corresponds a logical address to a physical address. **ARP accepts a logical address from the IP protocol, maps the address to the corresponding physical address** and pass it to the data link layer.

ARP Working

- ARP is used to map an **IPv4 address to a MAC address** within a local network.

Example:

Host A:

- IP: 192.168.1.10
- MAC: 00:AA:11:BB:22:CC

Host B:

- IP: 192.168.1.20
- MAC: 00:DD:33:EE:44:FF

Step 1: ARP Request (Broadcast)

- Host A sends:
- “Who has 192.168.1.20? Tell 192.168.1.10”
- Broadcast MAC: FF:FF:FF:FF:FF:FF

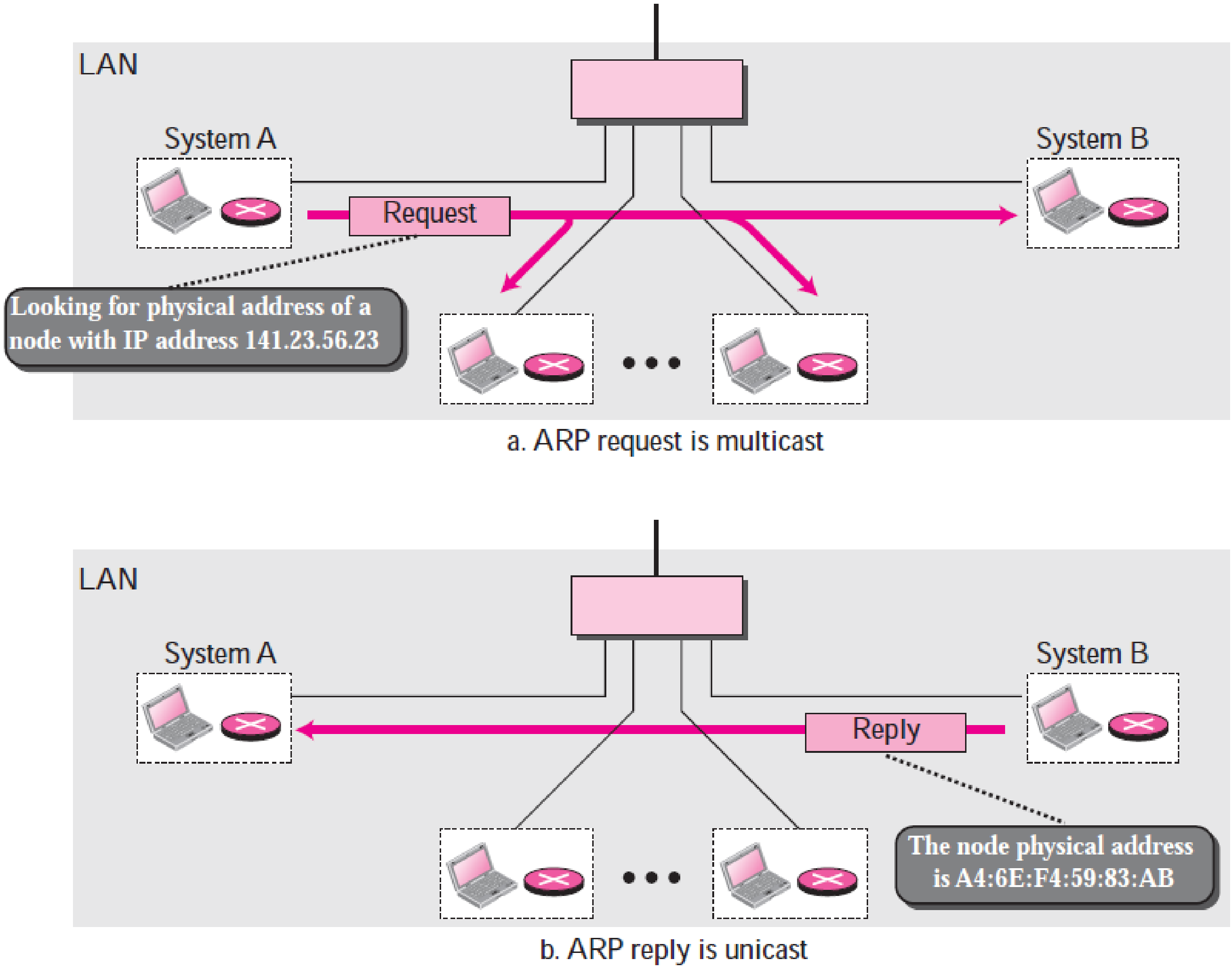
Step 2: ARP Reply (Unicast)

- Host B replies:
- “192.168.1.20 is at 00:DD:33:EE:44:FF”

Step 3: ARP Cache Update

- Host A stores mapping in ARP table.

ARP Operation



ARP Packet Format

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

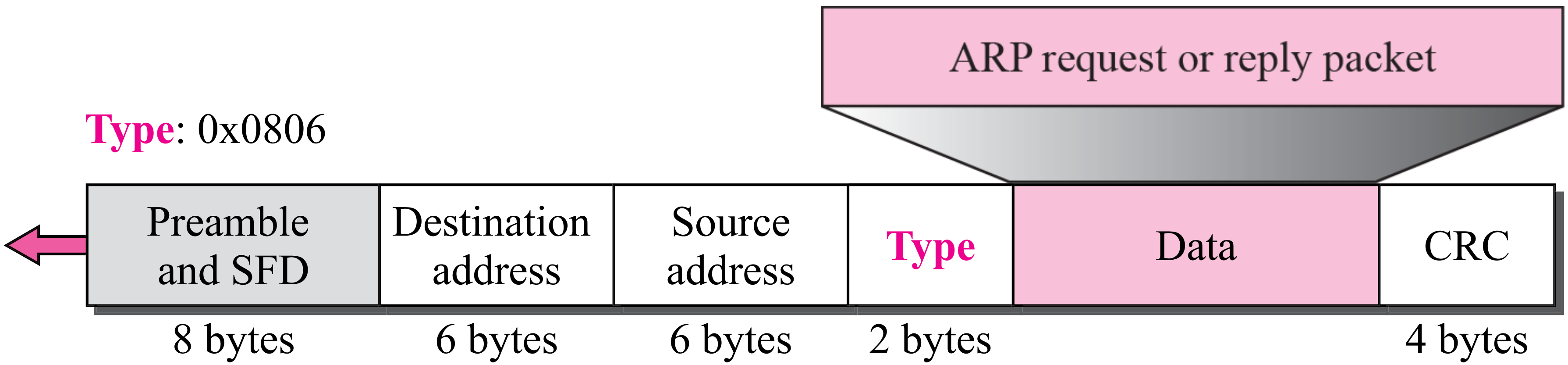
- **Hardware type.** This is a 16-bit field defining the type of the network on which ARP is running. Each LAN has been assigned an integer based on its type. For example, Ethernet is given the type 1. ARP can be used on any physical network.

- **Protocol type.** This is a 16-bit field defining the protocol. For example, the value of this field for the IPv4 protocol is 080016. ARP can be used with any higher-level protocol.

- **Hardware length.** This is an 8-bit field defining the length of the physical address in bytes. For example, for Ethernet the value is 6.

- **Protocol length**. This is an 8-bit field defining the length of the logical address in bytes. For example, for the IPv4 protocol the value is 4.

Encapsulation of ARP packet



Proxy ARP

- A technique called proxy ARP is used to create a subnetting effect.
- A proxy ARP is an ARP that acts on behalf of a set of hosts.
- Whenever a router running a proxy ARP receives an ARP request looking for the IP address of one of these hosts, the router sends an ARP reply announcing its own hardware (physical) address.
- After the router receives the actual IP packet, it sends the packet to the appropriate host or router.

Reverse Address Resolution Protocol (RARP)

- RARP finds the logical address for a machine that only knows its physical address.
- RARP requests are broadcast, RARP replies are unicast.
- Now obsolete (replaced by DHCP).

Working

- Example:
  Diskless machine with:
  MAC: 00:11:22:33:44:55

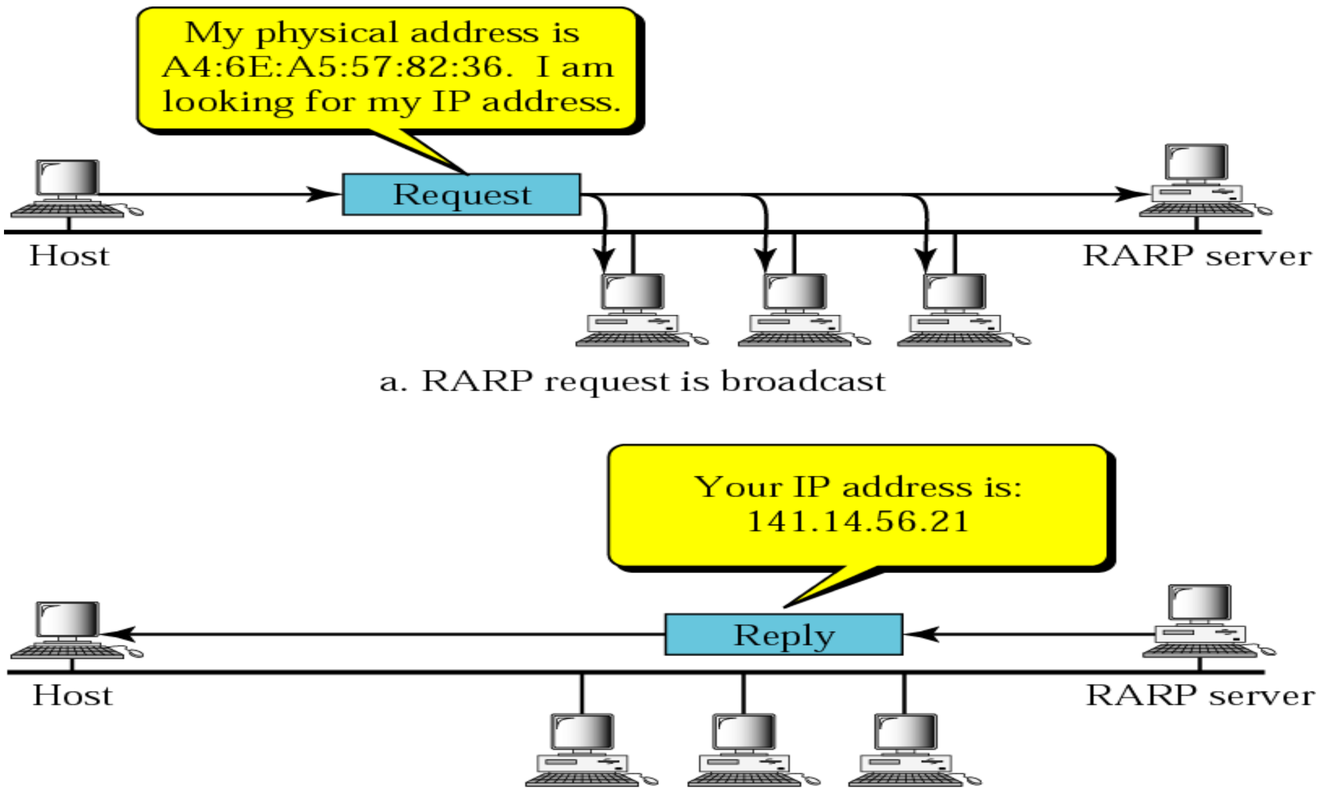
Step 1: RARP Request (Broadcast)

- “This is my MAC: 00:11:22:33:44:55. What is my IP?”

Step 2: RARP Reply (Unicast)

- Server responds:
- “Your IP is 192.168.1.50”

RARP Contd…



a. RARP request is broadcast

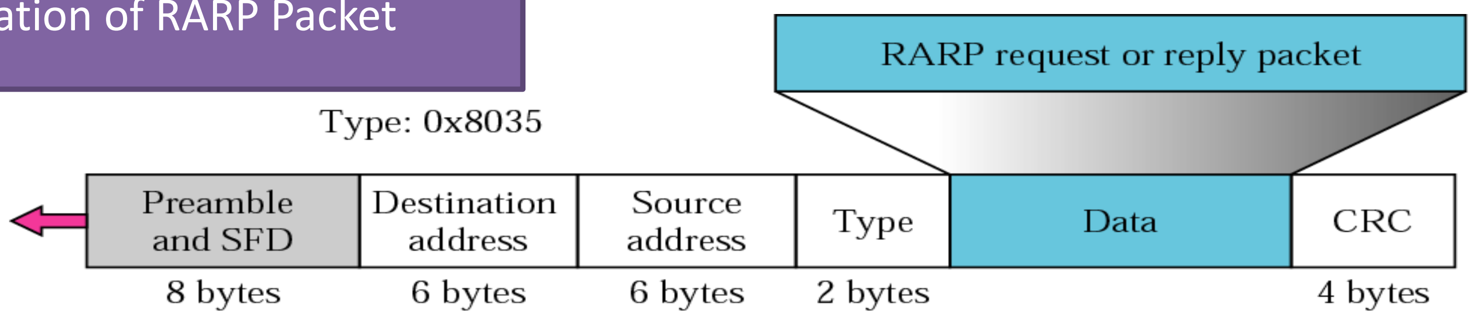
b. RARP reply is unicast

RARP Contd….

RARP Packet

Hardware type		Protocol type
Hardware length	Protocol length	Operation Request 3, Reply 4
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP) (It is not filled for request)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled for request)		
Target protocol address (For example, 4 bytes for IP) (It is not filled for request)		

Encapsulation of RARP Packet



BOOTP

- BOOTP (Bootstrap Protocol) is an application-layer protocol used to assign a **fixed IP address and boot information** to a client using its MAC address.

Core Concept

- Client knows only its **MAC address**
- Server maintains a **static MAC-to-IP mapping table**
- Server sends configuration information
- BOOTP provides static IP assignment.

Why BOOTP Was Needed?

- Early networks had **diskless workstations**.
- At startup, a diskless machine:
  - Does not know its IP address
  - Does not know subnet mask
  - Does not know default gateway
  - Does not know where to download OS from

  So it needed a server to provide:
  - IP address
  - Boot file name

  BOOTP solved this problem.

Step 1: BOOTP Request (Broadcast)

Client sends:

- “My MAC address is 00:AA:BB:CC:DD:EE. Please assign my IP address.”

Destination:

- IP: 255.255.255.255
- MAC: FF:FF:FF:FF:FF:FF

Step 2: BOOTP Reply

- Server checks its table.

Server replies:

- IP address: 192.168.1.50
- Subnet mask
- Default gateway
- Boot file name
- TFTP server address
- Client now boots successfully.

Limitations of BOOTP

- Not scalable
- Manual configuration required
- No dynamic IP assignment
- No automatic reuse of IP addresses

Because of these limitations, DHCP was introduced.

DHCP

- DHCP (Dynamic Host Configuration Protocol) is an application-layer protocol used to automatically assign IP addresses and other network configuration parameters to devices in a network.

DHCP

- DHCP is a network protocol that automatically gives devices an IP address and other network settings so they can connect and communicate on a network without manual configuration.
  - Assigns a unique IP address to each device.
  - Provides the subnet mask to define the local network range.
  - Configures the default gateway for communication outside the local network.
  - Supplies DNS server addresses for domain name resolution.

Why DHCP is Needed?

Problem in Large Networks

- Hundreds or thousands of devices
- Manual IP configuration is time-consuming
- Risk of IP conflicts
- Difficult management
- A protocol was needed for automatic IP assignment.

DHCP Basic Concept

- DHCP works using a **client-server model**.
- Client = Device requesting IP
- Server = Device assigning IP from pool
- Server maintains:
- IP Address Pool
  Lease Time
  Configuration Parameters

DHCP Components

- **DHCP Server:** The device (router/server) that manages the DHCP service, keeps the IP pool, and assigns IP addresses + configuration to clients.

- **DHCP Relay:** A router/switch feature that forwards DHCP requests and replies between clients and the server when they are in different networks (different subnets).

- **DHCP Client:** Any device (PC, mobile, printer, IoT) that requests an IP address and settings from the DHCP server.

DHCP Components

- **IP Address Pool:** The range of IP addresses available for allocation to clients (often with exclusions/reservations).

- **Lease:** The time period for which an assigned IP address is valid; the client must renew the lease to keep using it.

- **Default Gateway:** The router IP given to the client so it can communicate outside the local network (reach the internet/other networks).

- **DNS Servers:** DNS addresses provided to the client so it can convert domain names (like google.com) into IP addresses.

Operation Code	Hardware Type	Hardware Length	Hop Count
Transaction ID			
Number of Seconds		Flags	
Client IP Address			
Your IP Address			
Server IP Address			
Gateway IP Address			
Client Hardware Address (16 Byte)			
Server Name (64 Byte)			
Boot File Name (128 Byte)			
Option (Variable Length)			

DHCP Packet Format

- **Operation Code:**

1 → Boot Request (Client → Server)

2 → Boot Reply (Server → Client)

- **Hardware type: Ethernet(example)**
- **Hardware Length (8 bits):** Length of MAC address (e.g., 6 for Ethernet).
- **Hop Count (8 bits):** Maximum number of hops the packet can travel.
- **Transaction ID (32 bits):** Set by client, used to match requests and replies.
- **Number of Seconds (16 bits):** Time elapsed since the client started booting.
- **Flags (16 bits):** Leftmost bit indicates broadcast reply requirement.
- **Client IP Address (4 bytes):** Filled if the client already has an IP, else 0.

- **Your IP Address (4 bytes):** Client IP assigned by the server.
- **Server IP Address (4 bytes):** IP address of the responding DHCP server.
- **Gateway IP Address (4 bytes):** Router IP address (if applicable).
- **Client Hardware Address:** The device’s MAC address.
- **Server Name (64 bytes):** Optional server hostname.
- **Boot Filename (128 bytes):** Pathname of boot file (for diskless clients).
- **Options (variable):** Vendor-specific or additional configuration.

DHCP Working – DORA Process

- DORA
  - Discover
  - Offer
  - Request
  - Acknowledge
- This is the complete DHCP handshake.

What is DHCP Discover?

- It is the **first message** sent by a DHCP client to check if any DHCP server is available on the network.

When is it triggered?

- When a device boots up
- When it connects to a new network
- When it has no valid IP configuration

Transmission Type

- Broadcast
- Because the client does not know the DHCP server's IP address

Purpose

- To identify available DHCP servers
- To start the IP address assignment process

DHCP Offer Message

- A DHCP server replies to the Discover message with an Offer.

What does it include?

- Available IP address
- Subnet mask
- Default gateway
- DNS server
- Lease time
- Server Identifier

Important Technical Point

- The offered IP is placed in the **yiaddr field** (“Your IP Address” field in the DHCP header)

Transmission Type

- Broadcast or Unicast (depends on network behavior)

DHCP Request Message

- The client sends this message to:
- Accept one offered IP address
- Inform all other DHCP servers that their offers are rejected

When is it sent?

- After receiving one or more DHCP Offer messages.

Purpose

- Officially request the selected IP
- Prevent multiple servers from assigning different IPs to the same client
- This message is usually broadcast.

DHCP Acknowledgment (ACK)

- It is the final confirmation sent by the selected DHCP server.

Purpose

- Officially assign the IP address
- Confirm the lease time
- Provide complete network configuration
- After receiving DHCP ACK, the client:
  - Configures the IP address
  - Starts normal network communication

DHCP Negative Acknowledgment (NAK)

- It is sent by the server to reject the client’s request.

When is it sent?

- Requested IP is outside the configured scope
- IP is already assigned to another device
- Client has moved to a different network

Result

- The client must restart the DHCP process from Discover.

DHCP Decline

- Sent by the client when it detects that the offered IP address is already in use.

How does the client detect conflict?

Using:

- ARP probe
- Gratuitous ARP

Purpose

- Inform the server that the offered IP is unsafe

Result

- Server marks that IP as unavailable
- Another IP will be selected for future allocation

DHCP Release

- It allows the client to return the assigned IP address before lease expiration.

When does it occur?

- Device shuts down
- Device disconnects
- Interface disabled

Purpose

- Free the IP for reuse
- Improve IP address utilization

DHCP Inform

Used when:

- Client already has a manually configured (static) IP
- But needs additional network parameters

What does it request?

- DNS server
- Domain name
- NTP server
- Other configuration options

Transmission

- Usually unicast (client already has IP)

Message	Sent By	Purpose
Discover	Client	Find DHCP server
Offer	Server	Offer IP and configuration
Request	Client	Accept selected IP
ACK	Server	Confirm assignment
NAK	Server	Reject request
Decline	Client	Report IP conflict
Release	Client	Return IP to server
Inform	Client	Request configuration only

DHCP Advantages

- Automatic configuration
- Reduces manual errors
- Prevents IP conflicts
- Scalable
- Efficient IP reuse

Comparison

RARP	BOOTP	DHCP
Reverse Address Resolution Protocol	Bootstrap Protocol	Dynamic Host Configuration Protocol
Used to obtain IP address from MAC address	Used to assign IP and boot information	Used to dynamically assign IP and full network configuration
Works only within LAN	Works across networks (uses UDP/IP)	Works across networks (uses UDP/IP)
Requires dedicated RARP server	Requires BOOTP server	Uses DHCP server (usually router)
Static IP mapping	Static IP mapping	Dynamic IP allocation
Does not provide subnet mask, gateway, DNS	Provides subnet mask and gateway	Provides subnet mask, gateway, DNS, lease time
No lease mechanism	No lease mechanism	Uses lease-based allocation
Obsolete protocol	Rarely used	Widely used today
Replaced by BOOTP	Predecessor of DHCP	Most advanced and commonly used